

Batteryless Accelerometer Using Power Feeding System of RFID

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Abstract: The batteryless accelerometer which follows "ISO 15693 13.56 MHz radio frequency identification (RFID)" has developed with MEMS acceleration sensor which can detect $\pm 2G$ and has discrete circuit devices including micro processing unit (MPU) and thermistor. This module utilizes the radiowave for power feeding and signal communication as RFID. And therefore, the cables for power supply and signal transmission and the battery as a power supply are not required. The range of radiowave transmission is 1 cm and 3D antenna is utilized to emit and to perceive radiowave to and from any direction.

Keywords: RFID, ISO15693, 13.56 MHz, Accelerometer, 3D antenna

1. INTRODUCTION

Many kinds of wireless sensor network systems [2], [3], [4], [5], [6] have been studied, but most of the commercial products utilize battery as the power source[1].

Because of the troublesome battery exchange, it is hardly to bury under the ground in terms of handling. Furthermore, the deterioration may cause certain influence on environment when those were spread. In order to avoid those factors, the batteryless and cableless sensor module has been developed.

2. DESIGN OF WIRELESS SENSOR MODULE

2.1 Wireless Sensor Module

This wireless sensor module does not have battery, but instead, it is driven by power generation circuits which converts radiowave power to electric power like RFID. This specifications are shown in Table 1.

Table 1 Specifications of Wireless Sensor Module

items		specifications
power		electromagnetic induction
sensors	accelerometer	3-axis, -2G to 2G
	temperature	-10 to 60 degrees C
communication protocol		ISO15693
modulation scheme		ASK
radiowave frequency		13.56 MHz
distance of calls		10 mm
data transmission rate		26.48 kbps

The wireless sensor module consists of eight stacked 1 cm² ceramic boards. (Fig. 1) Each board is connected to the others by through-hole bus located on outer rim with solder.

Figure 2 shows circuit block, which consists of the antenna, interface, power generation, reception, transmission, sensor, multiplexer, and MPU embedded ADC.

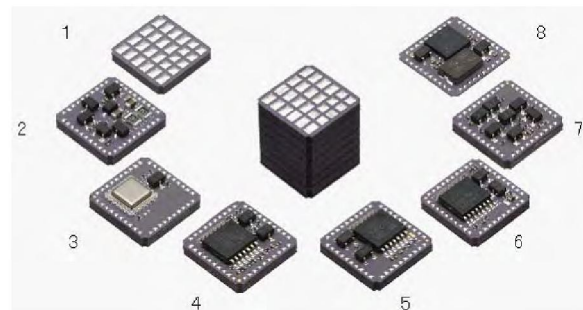


Fig. 1 Structure of Wireless Sensor Module.

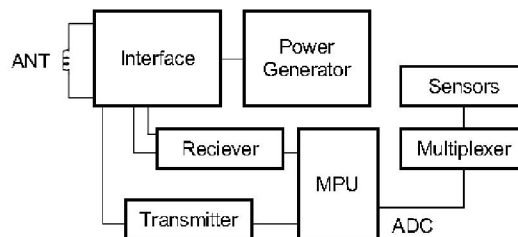


Fig. 2 Block diagram of Wireless Sensor Module.

The task and function of each layer of boards are as follows.

First layer (Top) : I/O terminals for writing program onto MPU and for monitoring signal of sensors. Most terminals of MPU are extracted to outside of the module for debug.

Second layer : Power generation and signal transmission circuits.

Third layer : 3-axis MEMS accelerometer.

Forth and fifth layers : Amplifying circuits.

Sixth : Multiplexer module for sensors output selection.

Seventh : Receiver circuits.

Eighth(Bottom) : MPU and a crystal oscillation circuit.

The moment the module receives radiowave from the host, electric power is generated by power circuit on the second layer, and the MPU on the eighth layer starts. The MPU observes radio signal, and receive *Inventory* signal.

After receiving, collision avoidance is processed within 320.9 μ s, and *Inventory* signal is transmitted by spontaneous one sensor. The read order from the host leads sensor measurement standby. The sensor is started by *Stay Quiet* instruction from the host, subsequently, the signal is converted by AD converter and transmitted. These process transitions were showed in Fig. 3.

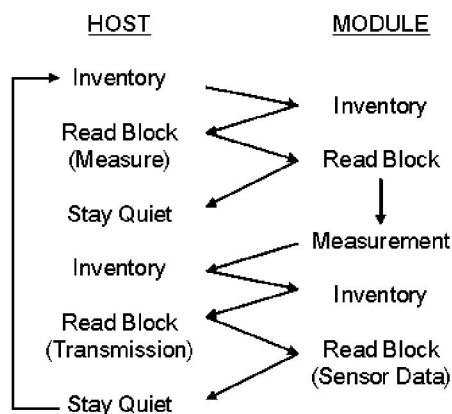


Fig. 3 State Diagram.

If the dice is rolled into the host antenna, 3D loop antenna receives radiowave, and electric power starts sensor module, and according to the routine depicted in Fig. 3, the sensor data is transmitted.

This module does not have the correction circuit. The volume does not exist, and even correction data is not inside MPU. The host stores the correction data and ID data obtained before shipment, corrects the sensor data with considering an ID of the module.

2.2 3-axis Accelerometer

We have developed a mass-producible sensor using a piezoresistive system and applying our proven MEMS technology and bipolar technology. The dimensions are 2.5 x 2.5 x 0.6 mm³. It is packaged with ceramic substrate of 4.8 x 4.8 x 1.3 mm³. And it has ± 2 G acceleration detection range and 500 Hz of frequency characteristics (Table 2).

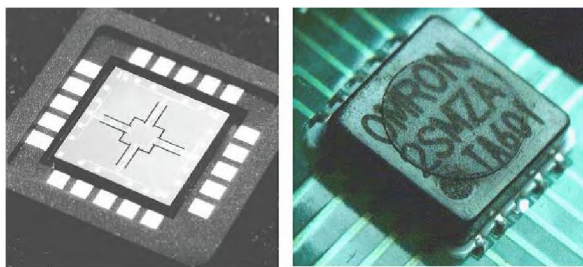


Fig. 4 3-axis accelerometer.

3. DESIGN OF 3D ANTENNA

The 2D loop antenna which was conventionally used in RFID had difficulty of radiowave communication from all three directions.

The left side of Fig. 5 depicts the simple application of 2D loop antenna for 3-axis. In this situation, although

only antenna parallel to the host antenna operates, 3 sets of electric circuit should be needed, and the scale of the circuit becomes too large.

Hence, the cube antenna structure which is made of one loop antenna was developed. This structure have the same characteristic from any direction of 3-axis. The 3D cubic antenna is shown in the right side of Fig. 5. This antenna has 2 side wires in each surface, and has total 8 sides. It has symmetry shape from all hexahedral surfaces and has ability of radiowave communication from any direction.

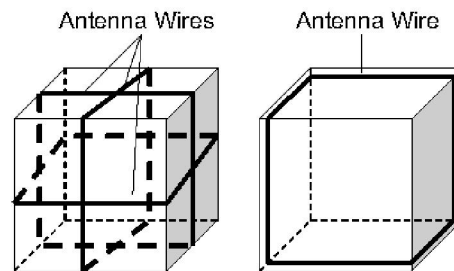


Fig. 5 Design of 3D Antenna

The antenna with 5 cm on a side and diameter of 0.5 mm is binded 6 times around the case to have magnetic permeability of 6 μ H. And it is soldered on the top layer of the sensor module.

4. MEASUREMENTS

4.1 Distance of Communication

The sensor module with the 3D antenna is shown in Fig. 6. The sensor module shown in Fig. 1 puts into the center of 5cm cubic dice. Both ends of 3D loop antenna is soldered to the antenna terminals on the top on module. It can make communication in the 1 cm distance between the bottom of the antenna and a host antenna.

The communication distance of the wireless sensor module with the 3D antenna is measured. For this measurement, the reader-writer, V720S-BC5D, and the host antenna, V720-HS03, both are manufactured by Omron Corporation are used. As the result, the distance was 1 cm. This distance is comparably short than the distance of conventional RFID, typically 100 cm. The reason of the difference is due to the difference of their power consumption, the conventional RFID is typically 0.2 mW and the wireless sensor module is 15 mW.

The high power consumption is induced at MPU. This wireless sensor module emulates the RFID function at MPU and at the high clock frequency of 13.56 MHz, the

Table 2 Characteristics of 3-axis Accelerometer

items	value	unit
piezoresistance	4.0	kohm
sensitivity	0.4	mV/V
offset	± 3.3	mV/V
linearity	± 1	%FS
frequency response	DC to 500	Hz

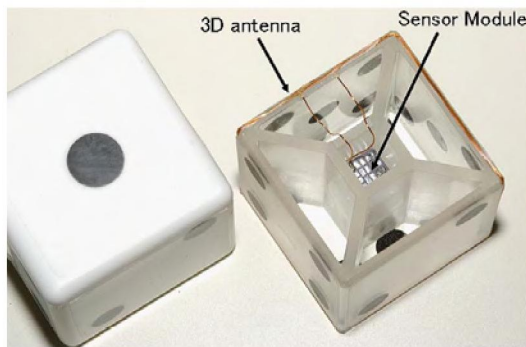


Fig. 6 Structure of Wireless Sensor Module and 3D Antenna for Dice.

same as the communication frequency of RFID, is used. The MPU has power consumption of 14.4 mW, about 96 % of total.

But, while the acceleration sensing by the MEMS sensor, the sensor's power consumption of 6 mW is needed to be secured, and therefore, the clock frequency is reduced to 6.78 MHz, a half of communication frequency. The frequency for MPU gets down half of communication mode in 6.78 MHz, because the MEMS accelerometer needs 6 mW when it is activated.

4.2 Exhibition

A simple game like poker using this dice was produced experimentally (Fig. 7).

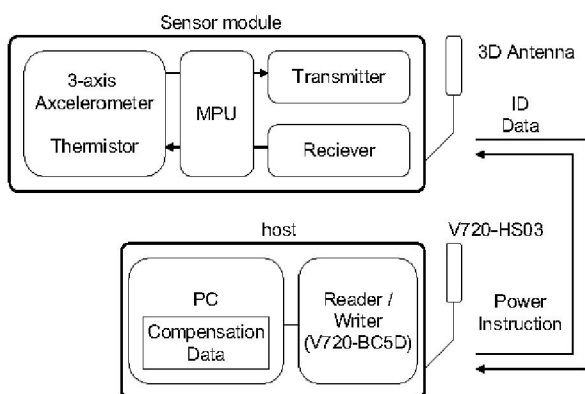


Fig. 7 System Architecture.

Each dice of multiple inside the host antenna transmits the ID and sensor data. The host stores the color, the ID, and the data related to the ID beforehand. The output from 3-axis accelerometer is associated with the spots on a dice like Fig. 8. The spots on a dice is recognized by the data corrected at the host. Fig. 9 depicts the display of the game software. 5 color dices is shown on the top part of the display, which is recognized by the output of 3-axis accelerometer shown below it. Because this module does not mount battery, battery exchange is needless, and very easy to treat.

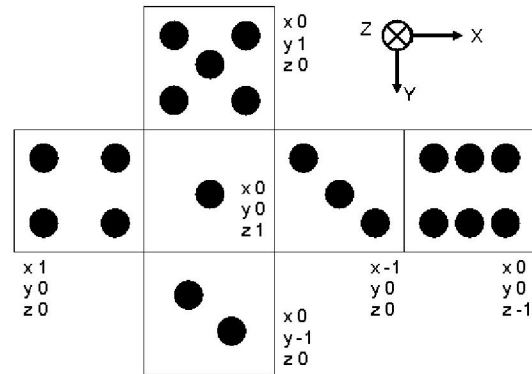


Fig. 8 The relation between the spots on a dice and acceleration.

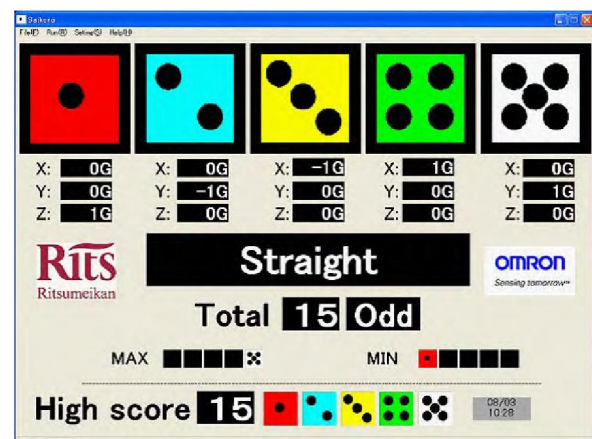


Fig. 9 The display example of game like poker.

5. CONCLUSIONS

The batteryless accelerometer which follows "ISO 15693 13.56MHz radio frequency identification (RFID)" is developed with a MEMS acceleration sensor and discrete circuit devices including MPU and a thermistor. The communication distance of the wireless sensor module was measured to be 1 cm. This short distance is due to the power consumption of the MPU.

In the future development, the MPU will be replaced by a custom LSI in order to reduce the power consumption by 1/100 and to realize the communication distance equivalent to the RFID.

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